

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

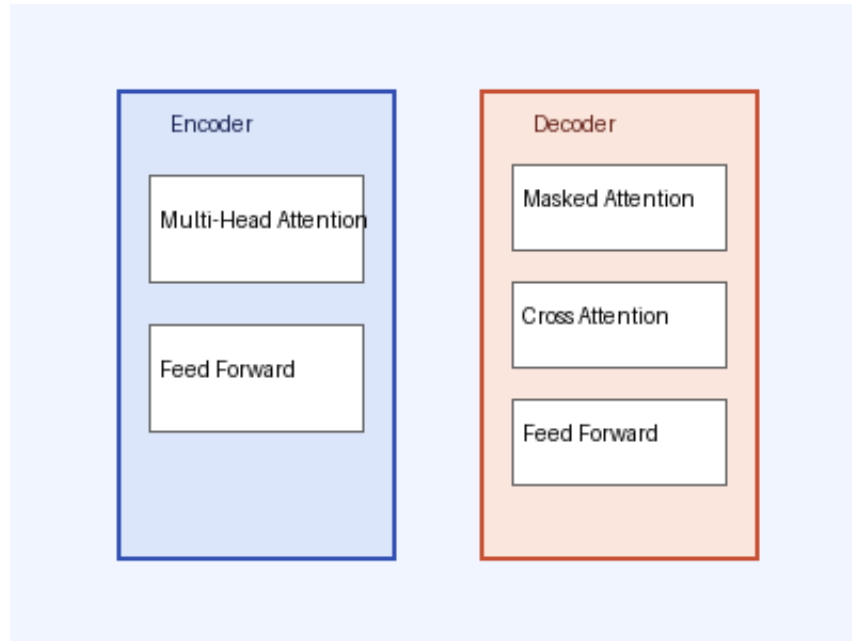
1. Introduction

Recurrent neural networks, long short-term memory and gated recurrent neural networks have been firmly established as state of the art approaches in sequence modeling. The fundamental constraint of sequential computation remains an obstacle to parallel training. In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output.

2. Model Architecture

The two main components of the Transformer architecture are the Encoder and the Decoder. The Encoder is composed of a stack of $N = 6$ identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism, and a simple, position-wise fully connected feed-forward network. The Decoder is also composed of a stack of $N = 6$ identical layers with an additional third sub-layer performing multi-head attention over the output of the encoder stack.

Figure 1: The Transformer - Model Architecture Diagram



3. Results and Evaluation

Model	BLEU (EN-DE)	BLEU (EN-FR)	Training Cost (FLOPs)
ByteNet	23.75	-	9.5e18
ConvS2S	25.16	40.46	9.6e18
Transformer (base model)	27.3	38.1	3.3e18
Transformer (big model)	28.4	41.8	2.3e19